

Module 1: Decision Under Uncertainty

ECO364-THINKING STRATEGICALLY

A single decision maker

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- ▶ In this module, we will consider a single decision maker making choices under uncertainty.
- ▶ Virtually, every decision is made in the face of uncertainty.
- ▶ Although in many of your previous ECON classes you didn't consider this aspect, many economic (and non economic!) problems require that we tackle uncertainty head on.
- ▶ In this module, we will consider the case of a single individual choosing some auction while facing uncertainty coming from "nature".
- ▶ Example: I am uncertain if it is **going to rain or not**. I **must chose** whether to take or not to take an umbrella. Of course, I would like to have an umbrella if it rains, otherwise I don't want to have one.
- ▶ You can think that the decision maker is "playing against nature".
- ▶ Things get more complicated **when interacting with another decision maker**. But this is the topic for all the other modules.
- ▶ If you haven't done so, check the syllabus for examples!

Some examples of non-intuitive decisions

- ▶ Prerequisite for this part: read the Review of Basic Probability from Prof. Learned-Miller, from the CS Department at the University of Massachusetts.
- ▶ It is a very concise review of some of the math that we will need for this part and the rest of the course!

The Monty Hall Problem

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About Let's Make a Deal

- ▶ Let's Make a Deal was a game show hosted by Monty Hall and Carol Merrill. It originally ran from 1963 to 1977 on network TV.
- ▶ The highlight of the show was the “Big Deal,” where contestants would trade previous winnings for the chance to choose one of three doors and take whatever was behind it--maybe a car, maybe livestock.
- ▶ Let's Make a Deal inspired a probability problem that can confuse and anger the best mathematicians, even Paul Erdős.

Suppose you're a contestant on Let's Make a Deal.

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You are asked to choose one of three doors.
The grand prize is behind one of the doors;
The other doors hide silly consolation gifts which Monty called “zonks”.
Let’s say that the grand prize is a brand new automobile.



You choose a door



Monty, who knows what's behind each of the doors, reveals a zonk behind one of the other doors.

He then gives you the option of switching doors or sticking with your original choice.

You get what is behind the door of your final choice.



To stick or switch?

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That is the question!

Does it matter?

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- ▶ From the participant's point of view, each door has equal likelihood of revealing the car
- ▶ Once the host opens an “empty” door, the car is behind one of the two remaining doors with equal likelihood
- ▶ Hence there is no advantage or disadvantage in changing the initial choice...
- ▶ right?

Please meet...

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Marilyn vos Savant

Columnist of the “Ask Marilyn” column in
Parade magazine

Listed in the Guinness Book of World Records for the “highest I.Q.” (228)

Marilyn's Claim

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- ▶ In her weekly column, she claimed that the participant would better switch doors
- ▶ She then received about 10,000 letters, the great majority disagreeing with her
- ▶ About 1,000 letters were written by mathematicians and scientists
- ▶ Even the New York Times published a long, front page article about this controversy in the July 21st, 1991, Sunday issue

A typical letter...

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This was written by Robert Sachs, a professor of mathematics at George Mason University:

“You blew it! Let me explain: If one door is shown to be a loser, that information changes the probability of either remaining choice –neither of which has any reason to be more likely– to $1/2$. As a professional mathematician, I am very concerned with the general public's lack of mathematical skills.

Please help by confessing your error and, in the future, being more careful.”

And like this letter, were many more, similarly from mathematicians and other professionals.

But Marilyn was right!

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There are two possible strategies:

1. To stick to the door which was chosen first
2. To switch, and open the other door

But Marilyn was right!

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- ▶ If the contestant chooses the “sticky” strategy her probability of winning the car is

$$1/3!$$

- ▶ If she chooses the “switchy” strategy she wins the car with probability ...

$$2/3!!$$

Why?

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- ▶ Suppose that she chooses to switch doors. If she had chosen first an empty door (which happens with probability $2/3$) she wins for sure: the host opens the other *empty* door and she then switches to the door with the car.

An alternative intuitive explanation

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- ▶ Suppose we change the rules of the game. Following the initial selection of a door, the contestant can either open it, or **open both** the remaining doors and win the car if it is behind one of them.
- ▶ It is obvious that now the contestant should choose to open the two remaining doors. **This way, the odds of winning double!**
- ▶ But this is exactly the situation in the original game. Except that there the host is helping the contestant by opening one of the two remaining doors for her – an empty one.

Let's see in depth what is going on in here...

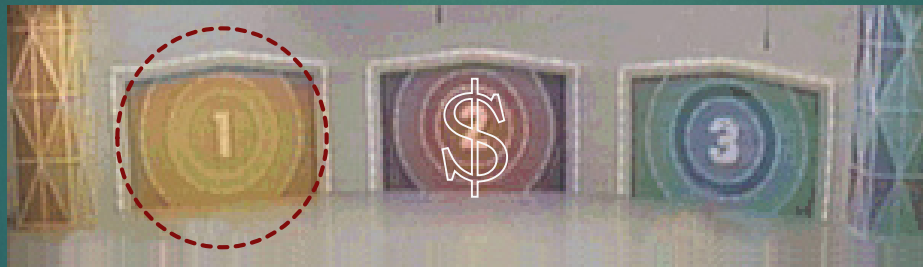
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- ▶ We have seen that given the option, [you should switch!](#)
- ▶ Assuming that Monty always gives you a chance to switch, you double your odds of winning by switching doors.
- ▶ We will see why, first by enumerating the possible cases, then by directly computing the probability of winning with each strategy.
- ▶ Last chance: read the notes on the Basic Review of probability because we are going to need it now!

Each door has a 1 in 3 chance of hiding the grand prize.

Suppose we begin by choosing door #1.

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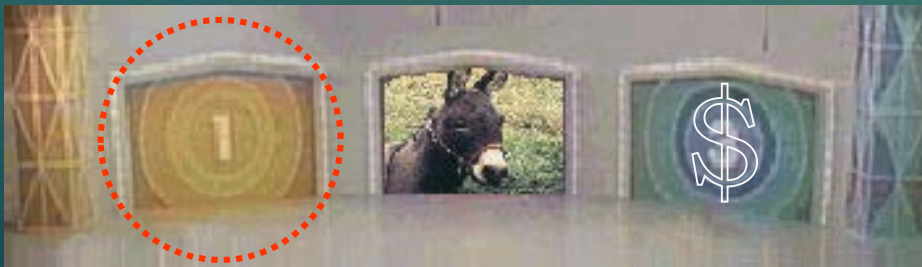
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In this case Monty may open either door #2 or #3



In both of these cases, Monty is forced to reveal the only other zongzi.



So what happens when you switch?

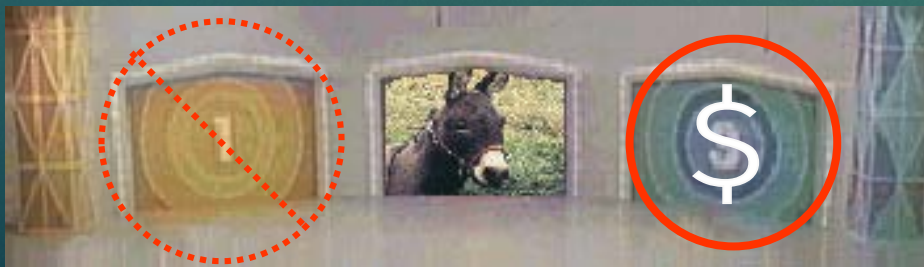
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In this case you were right the first time.
You lose!



In both of these cases,
you switch to the correct
door.
You win!



Let's try a little with another more “formal” method

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- ▶ To prove this result without listing all the cases, we need the notion of conditional probability.
- ▶ Conditional probability gives us a way to determine how the occurrence of one event affects the probability of another.
- ▶ Here, if we've chosen door #1 and Monty has opened door #2, we'd like to know the probability that the prize is behind door #1 and the probability that the prize is behind door #3 given this additional information.

Conditional probability

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- ▶ We can determine these probabilities using the rule

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- ▶ In words: The probability of event A given event B is the probability of both A and B divided by the probability of B.

In the following argument:

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- ▶ Assume that:
 - ▶ we originally chose door #1.
 - ▶ Monty opened door #2.
- ▶ Notation
 - ▶ Let “#1” denote the event that the prize is behind door #1, and similarly for doors #2 and #3.
 - ▶ Let “opened #2” denote the event that Monty has opened door #2.
- ▶ Our aim is to compute $p(\text{\#1} \mid \text{opened \#2})$ and $p(\text{\#3} \mid \text{opened \#2})$.

$$p(\#1 | \text{opened } \#2) = \frac{p(\#1 \cap \text{opened } \#2)}{p(\text{opened } \#2)}$$

$$p(\#3 | \text{opened } \#2) = \frac{p(\#3 \cap \text{opened } \#2)}{p(\text{opened } \#2)}$$

Rules :

$$1. p(A | B) = \frac{p(A \cap B)}{p(B)}$$

$$2. p(A \cap B) = p(B) \times p(A | B)$$

$$p(\#1 \cap \text{opened } \#2) = p(\text{opened } \#2 | \#1) \times p(\#1) \quad (\text{By rule 2.})$$

$$= \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

(If the prize is behind door #1, Monty can open either #2 or #3.)

$$p(\#3 \cap \text{opened } \#2) = p(\text{opened } \#2 | \#3) \times p(\#3) \quad (\text{By rule 2.})$$

$$= 1 \times \frac{1}{3} = \frac{1}{3}$$

(If the prize is behind door #3, Monty **must** open door #2.)

$$p(\#1 \mid \text{opened } \#2) = \frac{p(\#1 \cap \text{opened } \#2)}{p(\text{opened } \#2)} = \frac{1/6}{p(\text{opened } \#2)}$$

$$p(\#3 \mid \text{opened } \#2) = \frac{p(\#3 \cap \text{opened } \#2)}{p(\text{opened } \#2)} = \frac{1/3}{p(\text{opened } \#2)}$$

$$\begin{aligned} p(\text{opened } \#2) &= p(\text{opened } \#2 \cap \#1) + p(\text{opened } \#2 \cap \#2) + p(\text{opened } \#2 \cap \#3) \\ &= \frac{1}{6} + 0 + \frac{1}{3} = \frac{1}{2} \end{aligned}$$

So:

$$p(\#1 \mid \text{opened } \#2) = \frac{1/6}{1/2} = \frac{1}{3} \quad \text{and} \quad p(\#3 \mid \text{opened } \#2) = \frac{1/3}{1/2} = \frac{2}{3}$$

Conclusions:

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- ▶ Switching increases your chances of winning to $2/3$.
- ▶ A similar result holds for n doors.
- ▶ This strategy works only if we assume that Monty behaves predictably, offering a chance to switch every time.
- ▶ On Let's Make a Deal, Monty would play mind games with contestants, sometimes offering them money not to open the selected door.
- ▶ Play the game and check out the statistics at <http://math.ucsd.edu/~crypto/Monty/monty.html>
- ▶ Let's Make a Deal graphics courtesy of letsmakeadeal.com

Decision Making on Medical Diagnosis

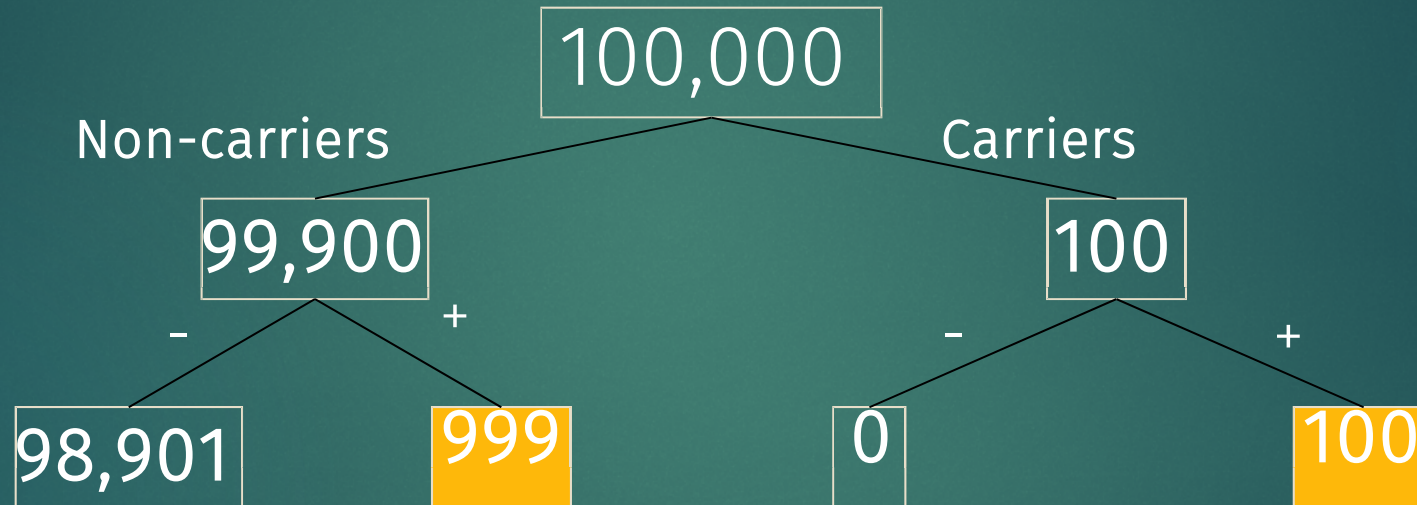
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- ▶ In a certain population one out of 1,000 people carries the HIV virus
- ▶ The test is 100% accurate on HIV carriers (every HIV carrier is diagnosed positive)
- ▶ The test is 99% accurate on non-HIV carriers (99 out of 100 non-carriers are diagnosed negative)
- ▶ A person chosen at random is tested positive
- ▶ What is the chance (probability) that he indeed carries the virus?

A surprising answer: less than 10%. Why?

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- ▶ Suppose for example that the population is 100,000 and all of them take the test



- ▶ Total number of “positive” tests is 1,099
- ▶ But only 100 of them carry the virus (the rest are false positives)
- ▶ Thus, the probability that a person tested positive indeed carries the virus is: $\frac{100}{1,099} \approx \frac{1}{11} \approx 9\%$

Let's redo the computation using Bayes Rule

- ▶ Let C (carrier) denote the event of being HIV positive. From the data we know that $\Pr(C)=0.001=0.1\%$. We also have that $\Pr(\text{Not } C)=0.999=99.99\%$
- ▶ Let POS denote the event that the HIV test comes positive.
 - ▶ Again, from the data we know that:
 - ▶ $\Pr(POS|C)=1=100\%$, $\Pr(\text{not } POS|C)=0$
 - ▶ $\Pr(\text{not } POS| \text{Not } C)=0.99=99\%$, $\Pr(POS|\text{not } C)=0.01=1\%$
- ▶ We need to calculate the probability that an individual chosen at Random has HIV given that he/she tested positive, i.e. $\Pr(C|POS)$
- ▶ By Bayes Rule

$$\begin{aligned}\Pr(C|POS) &= \frac{\Pr(POS|C) P(C)}{\Pr(POS|C) P(C) + \Pr(POS|\text{not } C) \Pr(\text{not } C)} = \frac{1 \times 0.001}{1 \times 0.001 + 0.01 \times 0.999} = \frac{0.001}{0.01099} \\ &= 0.0909 \approx 9\%\end{aligned}$$

- ▶ Again, if you haven't read, the notes on probability, this is section 2.5

On graphs and decisions

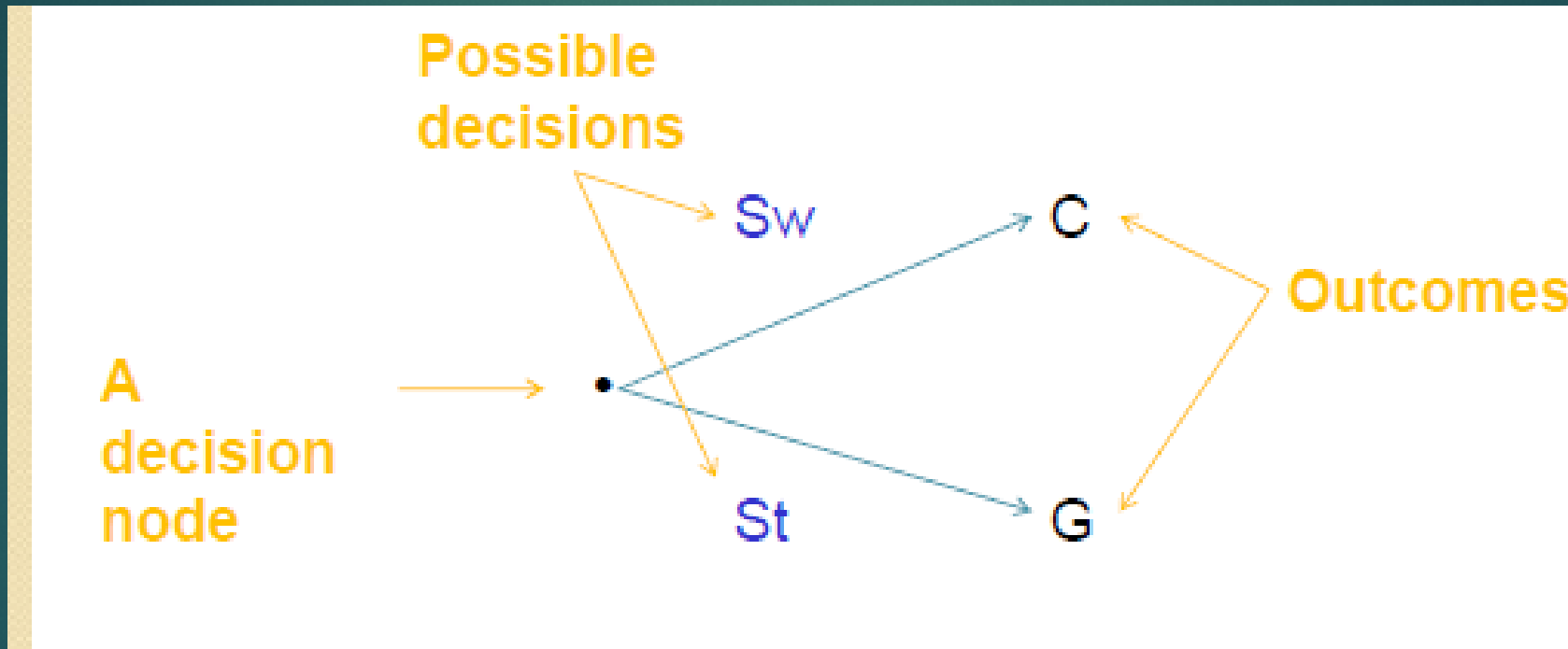
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- ▶ We've talked about a decision maker (player) making decisions
- ▶ Elements:
- ▶ Set of possible decisions (switch, stick)
- ▶ Possible outcomes (car, goat, lotteries)
- ▶ Preferences (car better than goat)
- ▶ And decisions may have to be made sequentially

On graphs and decisions

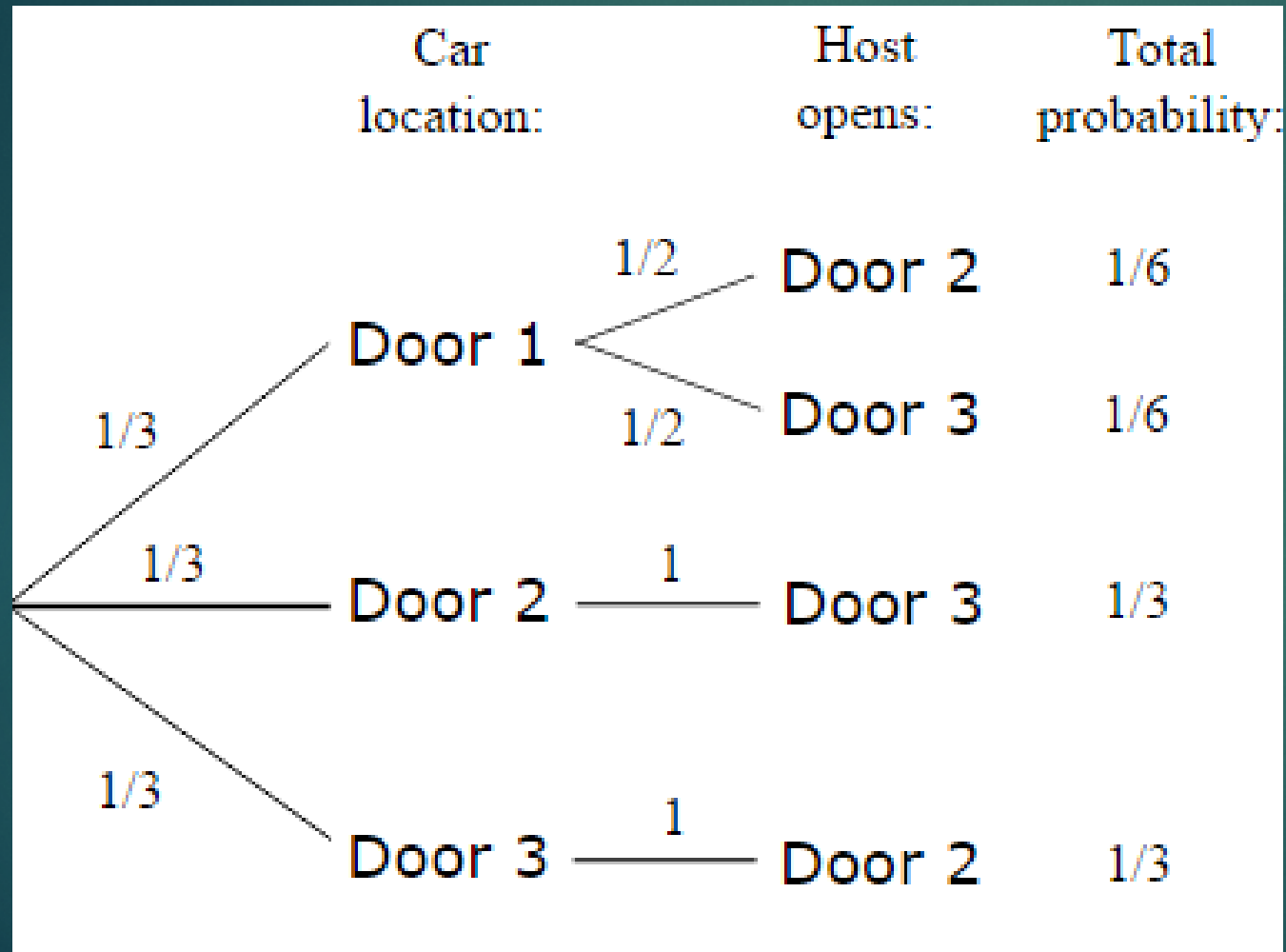
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- ▶ We use “decision trees” to represent these elements of the problem:



On graphs and decisions

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Stay

Switch

Car

Zonk

Car

Zonk

Zonk

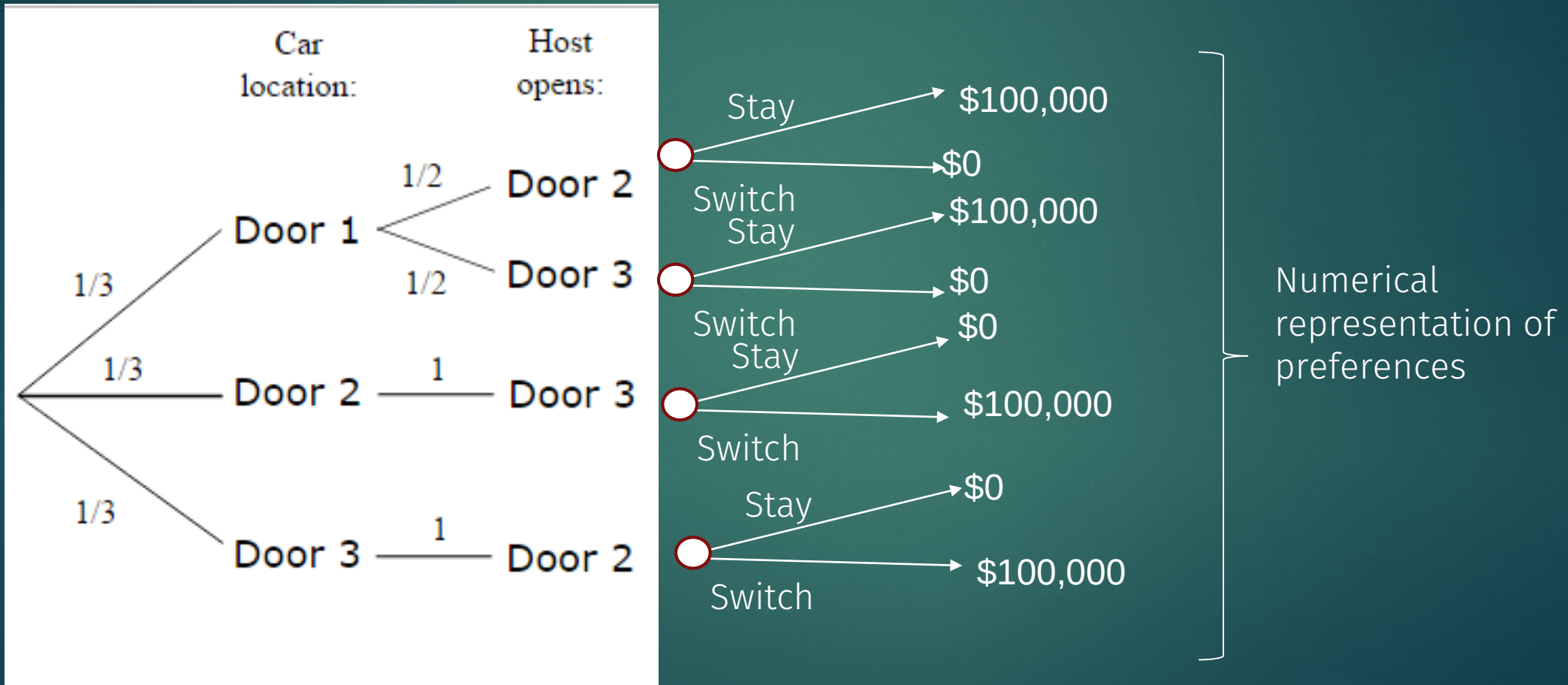
Car

Zonk

Car

On graphs and decisions

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Reading Assignment

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- ▶ I have posted Chapter 13 from the book An Introduction to Management Science, from Anderson, Sweeney, Williams, Camm and Martin.
- ▶ Read carefully section 13.1.
- ▶ You can skim section 13.2, but is not required.
- ▶ Read carefully section 13.3 since it deals with decision making with probabilities.
- ▶ Skim the rest of the chapter if you want.
- ▶ Section 13.5 will be useful later in the course, in module 4.